

Xiang Zhao

lesliezhaox@gmail.com | (+86) 18304049971 | <https://lesliezhao.github.io/>

EDUCATION

Northeastern University

Master of Engineering | Control Engineering

Core Courses: System Engineering and Decision Analysis, Applied Mathematics Statistics, Optimal Method and Theory, Computer Vision, Control System Design, Modern Control Engineering Design

Shenyang, China

2016.09 – 2019.01

Northeastern University

Bachelor of Engineering | Automation

Core Courses: Linear Algebra, Advanced Mathematics, Probability Theory and Mathematical Statistics, Complex Variable and Integral Transforms, Technology of Analog Electronics, Principles of Circuit, Technology of Mathematical Modeling, Machine Learning and Intelligent Optimization, Microcomputer Principle and Program Design, Electrical Machine and Applications, Process of Control System

Shenyang, China

2012.09 – 2016.06

RESEARCH EXPERIENCE

Research on Power Energy Prediction Based on Deep Learning LSTM Algorithm

Individual Researcher, Institute of Industrial and Systems Engineering, School of Information Science and Engineering (https://github.com/LeslieZhoa/Wind_predict_seq2seq)

Shenyang, China

2017.09-2018.09

- Designed an LSTM model and a Seq2Seq load forecasting model based on LSTM to address the power energy load issue of steel enterprises, and verified the effectiveness of the algorithm.
- Proposed two strategies to address the issue of missing historical data for wind power prediction: mean value completion and BP neural network completion.
- Combined the Seq2Seq prediction model based on LSTM with the wind power prediction task, bringing the attention mechanism into wind power prediction, significantly improving the prediction accuracy rate of the model by 1.3%, and achieving the state-of-the-art results.

WORK EXPERIENCE

Alibaba Hujing Digital Media & Entertainment Group

Senior Algorithm Engineer

Beijing, China

2023.08-Present

► Development of Ultra-realistic Digital Human Lip Shape Synthesis Technology

- Responsible for the entire process from data collection, processing, model selection, to algorithm development and real-time interaction logic programming, successfully developing a real-time lip movement driving technology at the leading level in China and achieving real-time voice-driven interaction in scenarios such as digital cabins.

► Development of Expression Editing and Video Expression Vividness Algorithm for Animation Production

- Developed expression editing and video vividness algorithms to enhance the animation production pipeline, improving 3D face-fitting accuracy and secondary video editing quality.
- Trained a model based on FLUX Kontext, first on paired rendered data and then fine-tuned on real faces with GRPO reinforcement learning. Designed dual rewards combining QwenVL-based emotion alignment and face ID similarity, boosting controllability and identity preservation.
- Built upon the WAN 2.2 framework, trained on anime and MEAD datasets, and refined via GRPO with emotion-feature constraints and perceptual loss, enhancing expression vividness while maintaining semantic consistency.

Microsoft Xiaoice

Senior Algorithm Engineer

Beijing, China

2022.06-2023.7

► Social Platform AI Virtual Photo Display

- Responsible for the development of the AI virtual idol image generation model, conducted LORA training for the AI virtual idol image generation network based on Stable Diffusion.
- Reproduced and optimized Heser (a head-swapping model based on GAN) and applied it to the group photos of real users and virtual idols on their social media.

► Virtual Human Face Rendering

- Involved in the algorithm development of virtual face replacement technology and optimized the face replacement algorithm based on HiFiFace.
- Adopted the multi-face recognition network fusion loss and innovatively designed the identity compensation network to compensate for the information loss in the pre-trained ArcFace features, resulting in a significant improvement in identity.
- Developed the FaceBlender model based on StyleGAN, using self-supervised learning to enhance the realism of the virtual person and solve the color difference problem after fusion.
(<https://github.com/LeslieZhoa/demos/blob/main/blend.png>)

AMAP-Alibaba Group

Senior Algorithm Engineer

Beijing, China

2021.08-2022.5

► Prediction and Optimization of Travel Time for Congested Road Sections in Self-Driving Vehicles

- Tasked with developing the algorithm for the precise prediction of congestion periods. The input features included the TopK similar road section features obtained through contrastive learning and historical data.
- Introduced the transformer to enhance the attention learning among features, revealing that the pass rate could be improved by 0.3%, and the failure rate decreased by about 15% compared with the previous period.

► Minimum Section Stratification

- Utilized contrastive learning to conduct the learning of minimum road section information in the feature space, supplemented additional inputs to the self-driving vehicle travel time prediction network, showing that the success rate increased by approximately 1% nationwide, while the failure rate decreased by 10% compared with the previous period.

BIGO Technology

Senior Algorithm Engineer

Beijing, China

2020.05-2021.08

► Development of the World's First Real-time Drivable Disney-style Face Sticker

- Spearheaded the development of the base large model and the distillation of a real-time, mobile-optimized small model.
- Handled post-processing rendering independently, successfully featured this innovative sticker, which gained widespread popularity in Southeast Asia.

► 'Photo Singing Sticker' Project

- Led super-resolution model training, development of the driving pipeline, and oversaw the overall project timeline, launching a special effect that was the first globally to support multi-person driving and achieving viral success in the Southeast Asian market.

TAL Education Group

Algorithm Engineer

Beijing, China

2019.01-2020.3

► Lip Shape Generation in the AI Classroom Scenario

- Accomplished the development of lip shape synthesis algorithms, establishing a mapping library from phonemes to Flame expression parameters.
- Projected the facial expression parameters obtained from the voice into the real human face as the condition input, trained the lip shape synthesis network based on StyleGAN, and finally implemented the lip shape synthesis in the offline AI classroom, achieving a leading domestic level of performance.

PUBLICATIONS

[1] Xinyi He, Jin Lang, Yanyan Zhang, Gengnan Zhao, and **Xiang Zhao**. (2022). Power Load Forecasting Method of Iron and Steel Enterprises based on Improved Long Short-term Memory Neural Network. Metallurgical Industry Automation, 46(3), 48-56. <https://doi.org/10.3969/j.issn.1000-7059>.

[2] **Xiang Zhao**, Yule Li, Wei Xiang. (2021). A Model Training, Image Processing Method, Device, Equipment and Medium. (CN 112767287 A).

[3] Lixin Tang, Jin Lang, and **Xiang Zhao**. (2018). A Deep Learning-based Ultra-short-term Wind Power Prediction Method. (CN 109615146 A).

[4] Yang Yang, Jin Lang, Jian Wu, Yanyan Zhang, and **Xiang Zhao**. (2023). A Novel Correlation-optimized Deep Learning Method for Wind Speed Forecast. Submitted.

[5] **Xiang Zhao**. (2018). Research on Power Energy Prediction Based on Deep Learning LSTM Algorithm. Submitted.

ACTIVITIES

<i>Director</i> , LearnHub Club, School of Northeastern University	2018.09-2019.06
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HONORS

<i>Third Prize in Fashion AI Global Challenge</i> , awarded to top 7% students by Alibaba Cloud Computing Co. Ltd.	2018.04
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<i>Second-class Scholarship</i> , awarded to top 10% students by Northeastern University	2013.03, 2013.09
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SKILLS

Computer Skills: C (6 yrs), C++ (6 yrs), Python (10 yrs), SQL (4 yrs), MATLAB (3 yr), LUA (2 yr), Microsoft Office

Language: Chinese (native), English (proficient)